



MLOPS & DATAOPS PIPELINE (TP CH2)

ETL Pipeline with Dagster, dbt and DuckDB



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G2

1. En quoi ce pipeline est-il plus structuré qu'un notebook unique ? :

Ce pipeline est plus structuré car il sépare les différentes étapes du traitement de données :

- ingestion (lecture des données)
- validation (contrôle qualité)
- transformation (dbt)
- tests (dbt test)

Chaque étape est indépendante dans un fichier dédié, ce qui améliore :

- la lisibilité
- la maintenance
- la réutilisation
- l'automatisation

⇒ Contrairement à un notebook, le pipeline est modulaire et adapté à la production.

2. Quel est le rôle de Git dans ce projet ?

Git permet :

- le versioning du code
- le suivi des modifications
- la collaboration
- le retour arrière (rollback)
- le déclenchement de la CI via GitHub Actions

⇒ Git est la base du MLOps moderne.

3. Quel est le rôle de requirements.txt ?

Le fichier requirements.txt sert à :

- lister toutes les dépendances Python
- garantir la reproductibilité de l'environnement
- permettre l'installation automatique dans CI/CD

Exemple :

- pandas

- duckdb
- dbt-core
- dagster

4. Pourquoi ajouter une étape validate.py ?

ette étape permet de :

- vérifier la qualité des données
 - détecter les valeurs nulles
 - vérifier le schéma des données
 - éviter de propager des erreurs dans le pipeline
- ⇒ Elle assure la data quality assurance.

5. Quel est le rôle de dbt test ?

dbt test permet de :

- tester la qualité des données après transformation
 - vérifier les contraintes (not null, unique)
 - valider la cohérence des modèles
- ⇒ Il garantit la fiabilité des données transformées.

6. Qu'apporte Dagster par rapport à une exécution manuelle ?

Dagster apporte :

- orchestration des étapes du pipeline
 - gestion des dépendances
 - visualisation du workflow
 - logs détaillés
 - possibilité de relancer une seule étape
 - monitoring centralisé
- ⇒ Il transforme des scripts en pipeline de production structuré.

7. Qu'apporte la CI ?

La CI (Continuous Integration) permet :

- l'exécution automatique du code à chaque push
 - la détection rapide des erreurs
 - la validation du pipeline avant production
 - l'amélioration de la qualité du code
- ⇒ Elle sécurise le développement et évite les erreurs en production.

8. Que manque-t-il encore pour un pipeline pleinement industrialisé ?

Le pipeline actuel n'est pas encore totalement industrialisé. Il manque :

- déploiement cloud (AWS, GCP, Azure)
- orchestration avancée (Dagster en production ou Airflow)
- monitoring et alerting (logs centralisés)
- containerisation (Docker)
- gestion des secrets (Vault / GitHub Secrets)
- base de données cloud (Snowflake, BigQuery)
- scheduling automatique en production
- CI/CD avec déploiement complet

=> Le projet actuel est un **prototype MLOps fonctionnel**, mais pas encore un système industriel complet.

✓ VERSION RÉELLE

GitHub Push

↓

GitHub Actions (CI/CD)

↓

ingest.py

↓

validate.py

↓

dbt run

↓

dbt test

↓

DuckDB

✓ DAGSTER À PART

Dagster UI (local)

↓

Launch Run manuel

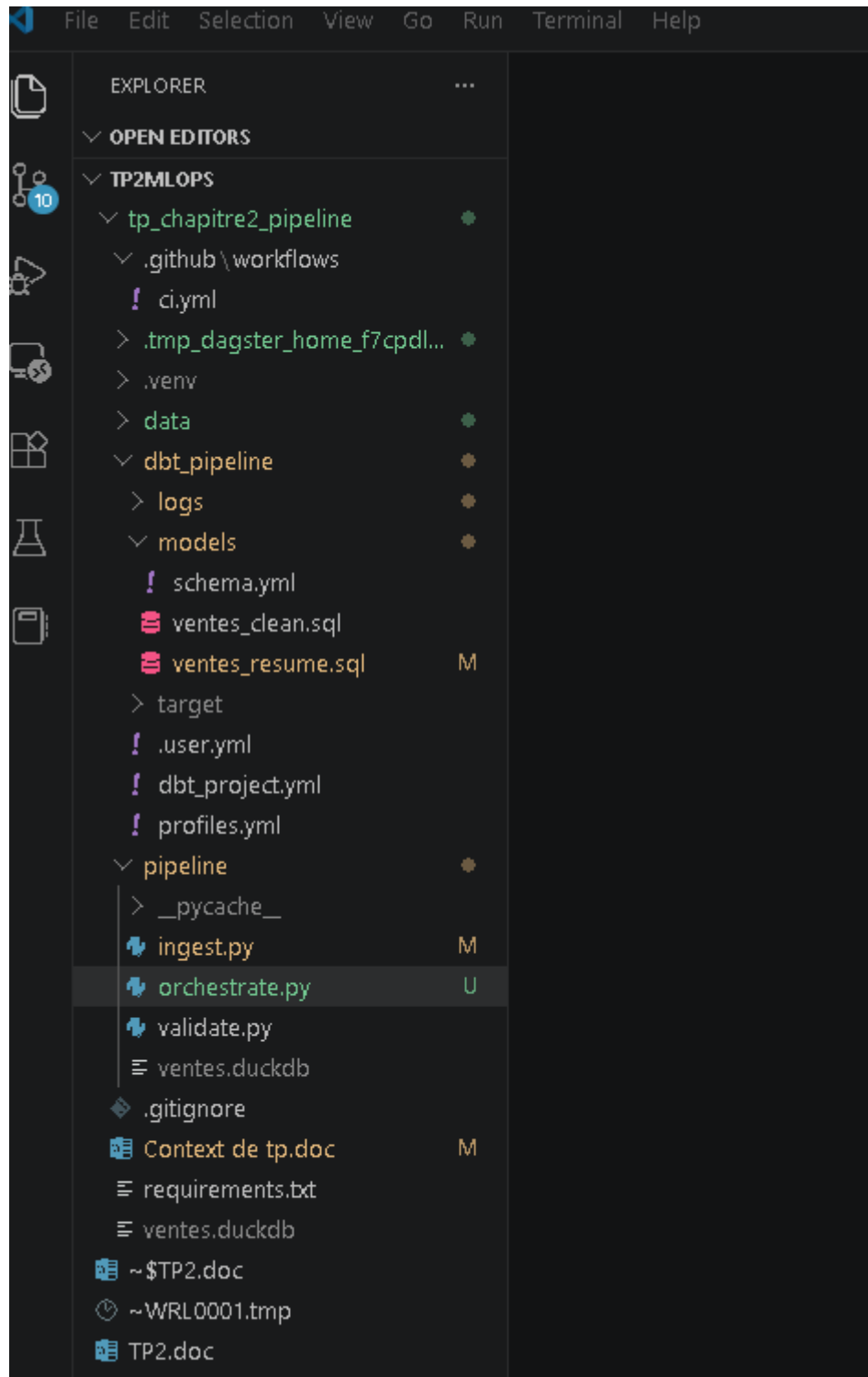
↓

observe ingest / validate / dbt / test

↓

logs + monitoring

SCREEN 1 — STRUCTURE DU PROJET



2. SCREEN 2 — GIT INIT + COMMIT

```
C:\Windows\System32\cmd.exe

:\Users\HP\Desktop\tp2mlops>cd tp_chapitre2_pipeline

:\Users\HP\Desktop\tp2mlops\tp_chapitre2_pipeline>git status
On branch master
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   Context de tp.doc
    modified:   dbt_pipeline/logs/dbt.log
    modified:   dbt_pipeline/models/ventes_resume.sql
    modified:   pipeline/ingest.py
    deleted:    ~$ntext de tp.doc

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .tmp_dagster_home_f7cpdljd/
    data/
    pipeline/orchestrate.py

no changes added to commit (use "git add" and/or "git commit -a")

:\Users\HP\Desktop\tp2mlops\tp_chapitre2_pipeline>
```

```
no changes added to commit (use "git add" and/or "git commit -a")
```

```
C:\Users\HP\Desktop\tp2mlops\tp_chapitre2_pipeline>git log
commit 0a896991da244b1e296074f4e3b4c33c826003d6 (HEAD -> master)
Author: mohamed-makrani <makrani.mohamed@ump.ac.ma>
Date: Mon May 25 20:38:37 2026 +0000
```

Ajout d'une CI simple

```
commit 4937497f2de0d5e652ccab9f67255a761e37d4bd
Author: mohamed-makrani <makrani.mohamed@ump.ac.ma>
Date: Mon May 25 20:12:02 2026 +0000
```

Ajout des tests simples dbt

```
commit a8ce17e19a63b39b9ba9891ceda418ae8f72fa50
Author: mohamed-makrani <makrani.mohamed@ump.ac.ma>
Date: Mon May 25 20:09:49 2026 +0000
```

Ajout des modèles dbt

```
commit d594237c686d3a8e7c49c6717479ac6189ac561f
Author: mohamed-makrani <makrani.mohamed@ump.ac.ma>
Date: Mon May 25 19:51:19 2026 +0000
```

ajout de l'etape de validation minimale

```
commit 88a369cbe9b525d734e5ab96859ebf48924c44d9
Author: mohamed-makrani <makrani.mohamed@ump.ac.ma>
Date: Mon May 25 17:47:30 2026 +0000
```

```
initialisation du project pipeline
:...skipping...
commit 0a896991da244b1e296074f4e3b4c33c826003d6 (HEAD -> master)
Author: mohamed-makrani <makrani.mohamed@ump.ac.ma>
Date: Mon May 25 20:38:37 2026 +0000
```

Ajout d'une CI simple

```
commit 4937497f2de0d5e652ccab9f67255a761e37d4bd
Author: mohamed-makrani <makrani.mohamed@ump.ac.ma>
Date: Mon May 25 20:12:02 2026 +0000
```

Ajout des tests simples dbt

```
commit a8ce17e19a63b39b9ba9891ceda418ae8f72fa50
Author: mohamed-makrani <makrani.mohamed@ump.ac.ma>
Date: Mon May 25 20:09:49 2026 +0000
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Ajout des modèles dbt

```
commit d594237c686d3a8e7c49c6717479ac6189ac561f
Author: mohamed-makrani <makrani.mohamed@ump.ac.ma>
Date: Mon May 25 19:51:19 2026 +0000
```

ajout de l'etape de validation minimale

```
commit 88a369cbe9b525d734e5ab96859ebf48924c44d9
Author: mohamed-makrani <makrani.mohamed@ump.ac.ma>
```


3. SCREEN — TERMINAL DAGSTER START

```
venv) C:\Users\HP\Desktop\tp2mlops\tp_chapitre2_pipeline>dagster dev -f pipeline\orchestrate.py
26-05-25 20:40:43 +0000 - dagster - INFO - Using temporary directory C:\Users\HP\Desktop\tp2mlops\tp_chapitre2_pipeline\tmp_dagster_h
f7cpdljd for storage. This will be removed when dagster dev exits.
26-05-25 20:40:43 +0000 - dagster - INFO - To persist information across sessions, set the environment variable DAGSTER_HOME to a dire
y to use.
26-05-25 20:40:46 +0000 - dagster - INFO - Launching Dagster services...
26-05-25 20:41:02 +0000 - dagster.daemon - INFO - Instance is configured with the following daemons: ['AssetDaemon', 'BackfillDaemon',
eshnessDaemon', 'QueuedRunCoordinatorDaemon', 'SchedulerDaemon', 'SensorDaemon']
26-05-25 20:41:05 +0000 - dagster - WARNING - C:\Users\HP\Desktop\tp2mlops\tp_chapitre2_pipeline\.venv\lib\site-packages\dagster\core
cution\compute_logs.py:53: UserWarning: WARNING: Compute log capture is disabled for the current environment. Set the environment vari
`PYTHONLEGACYWINDOWSTDIO` to enable.

warnings.warn(WIN_PY36_COMPUTE_LOG_DISABLED_MSG)

26-05-25 20:41:05 +0000 - dagster-webserver - INFO - Serving dagster-webserver on http://127.0.0.1:3000 in process 12992
26-05-25 21:32:15 +0000 - dagster - INFO - KeyboardInterrupt received
26-05-25 21:32:15 +0000 - dagster - INFO - Shutting down Dagster services...

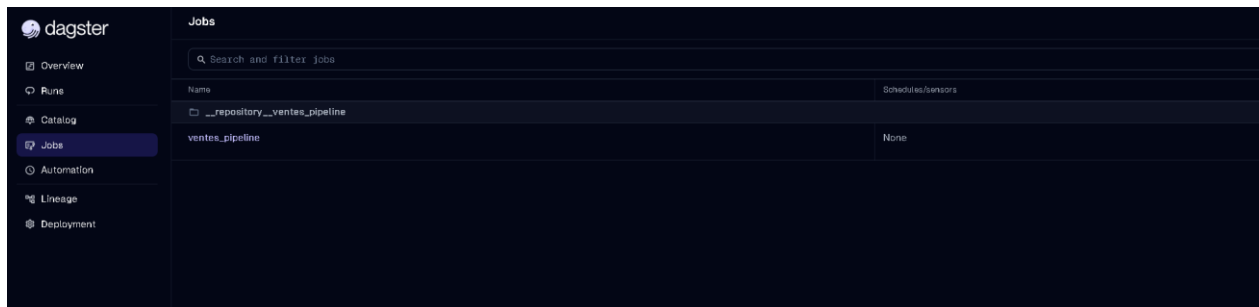
orted!
26-05-25 21:32:18 +0000 - dagster.daemon - INFO - Received interrupt, shutting down daemon threads...
26-05-25 21:32:18 +0000 - dagster.daemon - INFO - Daemon threads shut down.

venv) C:\Users\HP\Desktop\tp2mlops\tp_chapitre2_pipeline>dagster dev -f pipeline\orchestrate.py
26-05-25 21:41:11 +0000 - dagster - INFO - Using temporary directory C:\Users\HP\Desktop\tp2mlops\tp_chapitre2_pipeline\tmp_dagster_h
7mx1i_s7 for storage. This will be removed when dagster dev exits.
26-05-25 21:41:11 +0000 - dagster - INFO - To persist information across sessions, set the environment variable DAGSTER_HOME to a dire
y to use.
26-05-25 21:41:15 +0000 - dagster - INFO - Launching Dagster services...
26-05-25 21:41:31 +0000 - dagster.daemon - INFO - Instance is configured with the following daemons: ['AssetDaemon', 'BackfillDaemon',
eshnessDaemon', 'QueuedRunCoordinatorDaemon', 'SchedulerDaemon', 'SensorDaemon']
26-05-25 21:41:33 +0000 - dagster - WARNING - C:\Users\HP\Desktop\tp2mlops\tp_chapitre2_pipeline\.venv\lib\site-packages\dagster\core
cution\compute_logs.py:53: UserWarning: WARNING: Compute log capture is disabled for the current environment. Set the environment vari
`PYTHONLEGACYWINDOWSTDIO` to enable.

warnings.warn(WIN_PY36_COMPUTE_LOG_DISABLED_MSG)

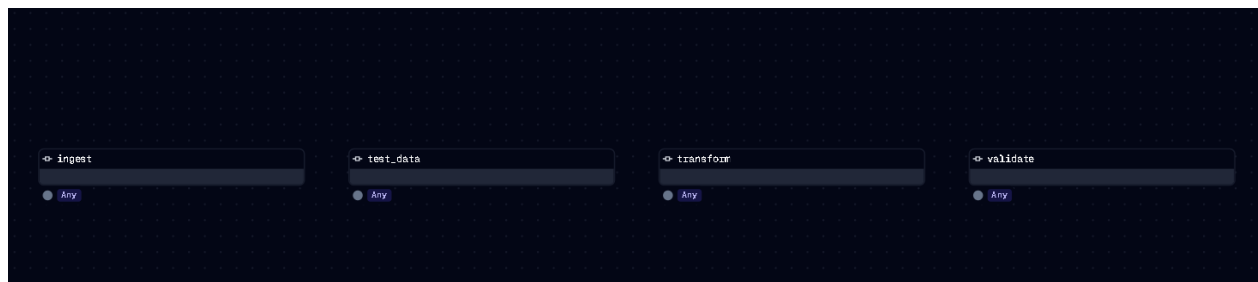
26-05-25 21:41:33 +0000 - dagster-webserver - INFO - Serving dagster-webserver on http://127.0.0.1:3000 in process 5096
```

4. SCREEN — DAGSTER HOME PAGE



Name	Schedules/sensors
repository__ventas_pipeline	
ventas_pipeline	None

5. SCREEN — GRAPH DU PIPELINE



6. SCREEN — LAUNCH RUN

The screenshot displays the Launchpad interface for a run titled '4670dfb4'. The run is marked as 'Success' and occurred on May 25, 9:44:40 PM. The interface shows a progress bar at the top with a 'Hide not started steps' checkbox. Below the progress bar, a list of steps is shown: 'test_data', 'transform', 'validate', and 'ingest'. Each step has a corresponding bar indicating its duration. A search bar and a 'Hide unselected steps' checkbox are located below the steps list. On the right side, there is a panel with status indicators for 'Preparing (0)', 'Executing (0)', 'Errored (0)', and 'Succeeded (4)'. Below this panel, a table of events is displayed, showing the timeline of the run.

TIME/STAMP	OP	EVENT TYPE	INFO
01:45:05.830 PM	ingest	STEP_OUTPUT	Yielded output "result" of type "Any". (Type check passed).
01:45:08.254 PM	ingest	HANDLED_OUTPUT	Handled output "result" using IO manager "io_manager"
01:45:09.692 PM	ingest	STEP_SUCCESS	Finished execution of step "ingest" in 10.9s.
01:45:09.733 PM	validate	STEP_OUTPUT	Yielded output "result" of type "Any". (Type check passed).
01:45:12.886 PM	validate	HANDLED_OUTPUT	Handled output "result" using IO manager "io_manager"
01:45:14.643 PM	validate	STEP_SUCCESS	Finished execution of step "validate" in 21.87s.
01:45:15.959 PM	transform	STEP_OUTPUT	Yielded output "result" of type "Any". (Type check passed).
01:45:15.872 PM	transform	HANDLED_OUTPUT	Handled output "result" using IO manager "io_manager"
01:45:15.927 PM	test_data	STEP_OUTPUT	Yielded output "result" of type "Any". (Type check passed).
01:45:15.938 PM	transform	STEP_SUCCESS	Finished execution of step "transform" in 1m8s.
01:45:15.183 PM	test_data	HANDLED_OUTPUT	Handled output "result" using IO manager "io_manager"
01:45:15.238 PM	test_data	STEP_SUCCESS	Finished execution of step "test_data" in 1m9s.
01:46:02.960 PM	-	ENGINE_EVENT	Multiprocess executor: parent process exiting after 1m20s (pid: 18376) pid: 18375
01:46:03.802 PM	-	RUN_SUCCESS	Finished execution of run for "ventas_pipeline".
01:46:03.472 PM	-	ENGINE_EVENT	Process for run exited (pid: 18376).